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S. H. TUTORIALS



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REFERENCE BOOKS

- ❖ Digital Systems, Principle & Applications, Ronald J. Tocci, Prentice-Hall
- ❖ Digital Design, M. Morris Mano, Michael D. Ciletti, Pearson Education, Inc.
- ❖ Digital Circuits and Design, S. Salivahanan, S. Arivazhagan, Oxford University Press.
- ❖ Digital Electronics, G. K. Kharate, Oxford University Press.
- ❖ Digital Electronics, Bignell James, Logic and Systems, Cengage Learning
- ❖ Digital Logic and Computer Design, M. Morris Mano, Pearson Education, Inc.

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- ❖ A logic gate is an electronic circuit which makes logic decisions.
- ❖ The most common logic gates are **OR, AND, NOT, NAND** and **NOR** gates.
- ❖ The **NAND** and **NOR** gates are called Universal gates.
- ❖ Logic gates have two or more inputs and only one output except for the NOT gate which has only one input.

OR GATE: The OR gate performs logical addition, known as OR function.

The OR gate has two or more inputs. The operation OR gate is such that a



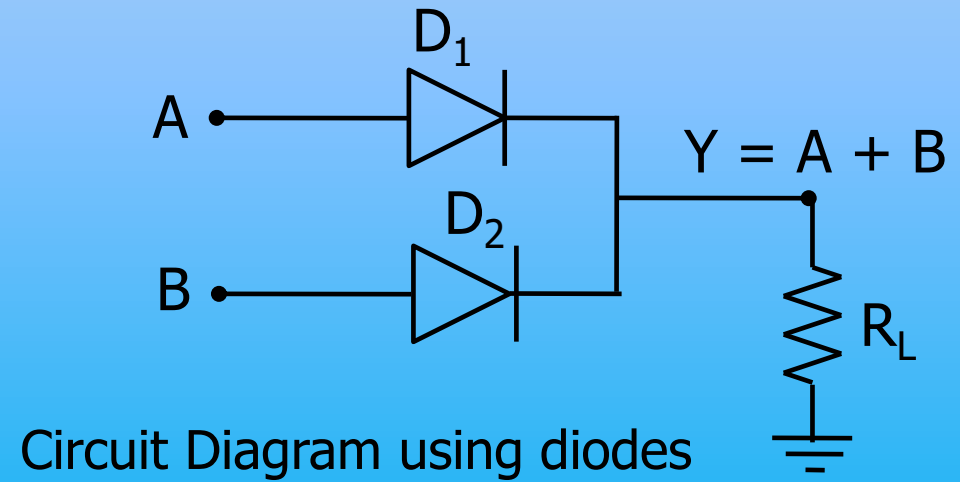
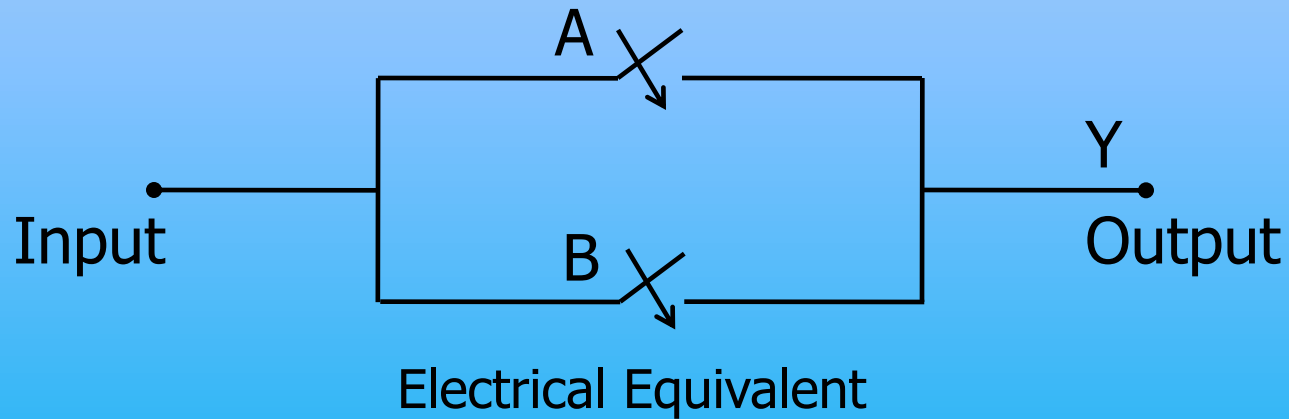


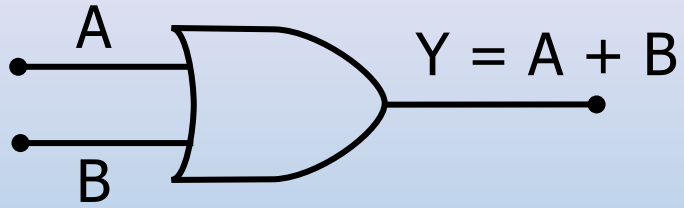
HIGH (1) on the input is produced when any of the inputs is HIGH(1). The output is LOW (0) only when all the inputs are LOW (0).

If A and B are the input variables of an OR gate and Y is its output, then

$$Y = A + B$$

Similarly, for more than two variables $Y = A + B + C + \dots$





Symbol

TRUTH TABLE

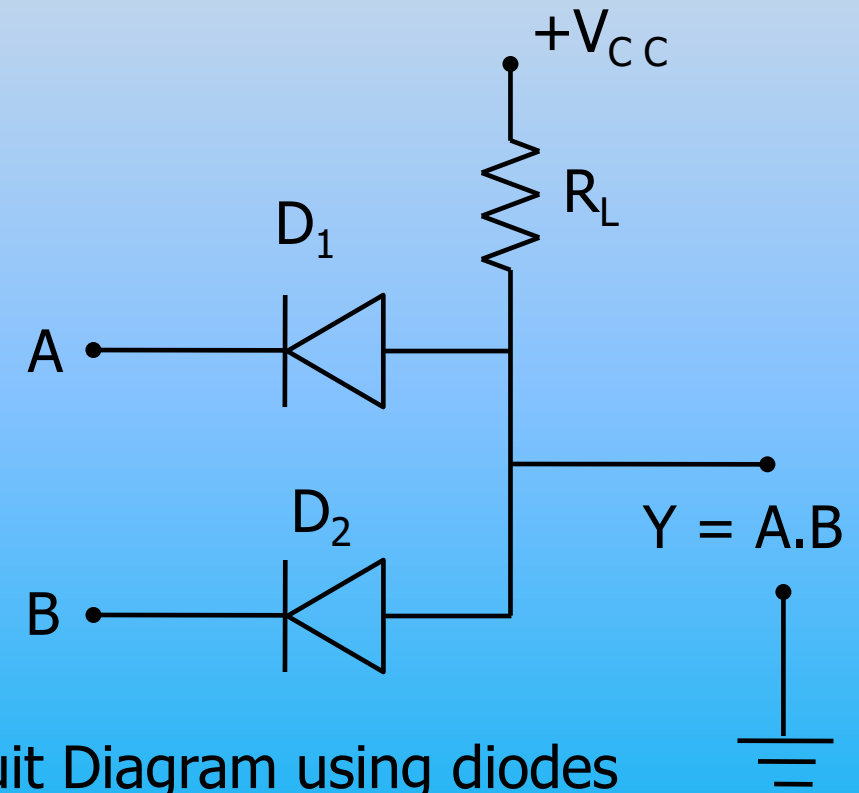
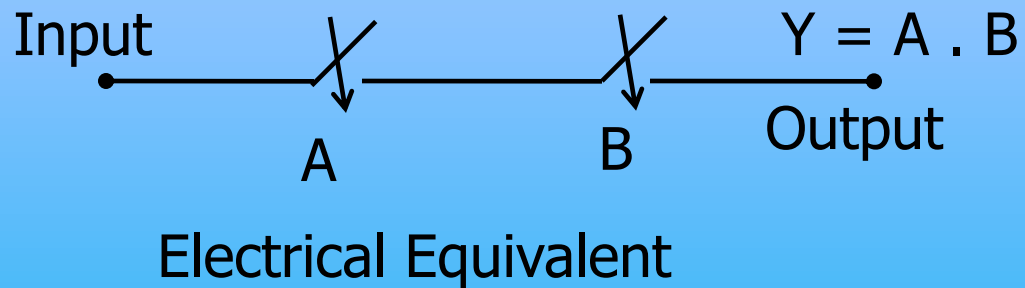
INPUTS		OUTPUT
A	B	
0	0	0
0	1	1
1	0	1
1	1	1



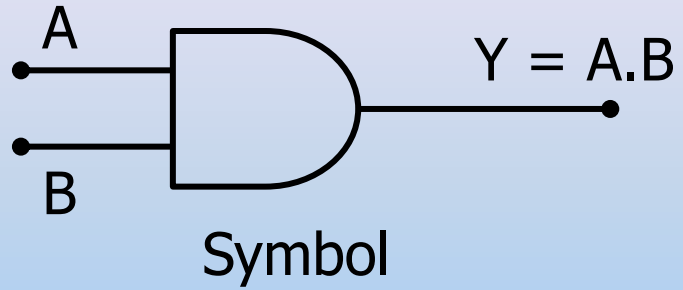
AND Gate : The AND gate performs logical multiplication, commonly known as AND function. The AND gate has two or more inputs and single output.

If A and B are input variables and Y is the output then

$$Y = A \cdot B$$



Circuit Diagram using diodes

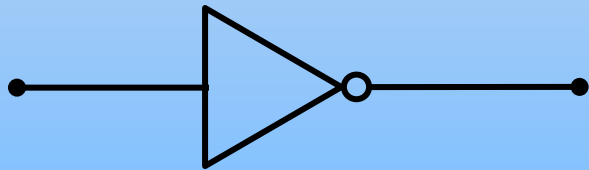


TRUTH TABLE

INPUTS		OUTPUT
A	B	
0	0	0
0	1	0
1	0	0
1	1	1



NOT Gate (Inverter) : The NOT gate performs the basic logical function called inversion or complementation. The purpose of this gate is to convert one logic level into the opposite logic level. It has one input and one output. When a HIGH level is applied to an inverter, a LOW level appears at its output and vice versa.



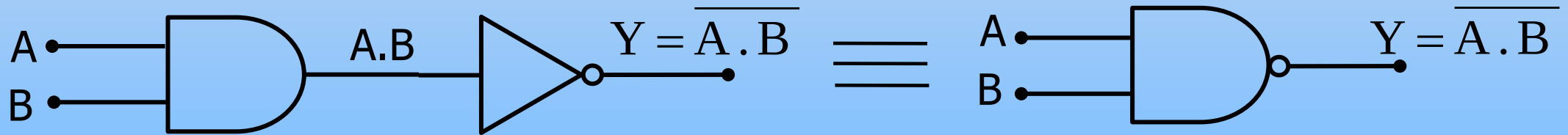
Symbol

Truth Table

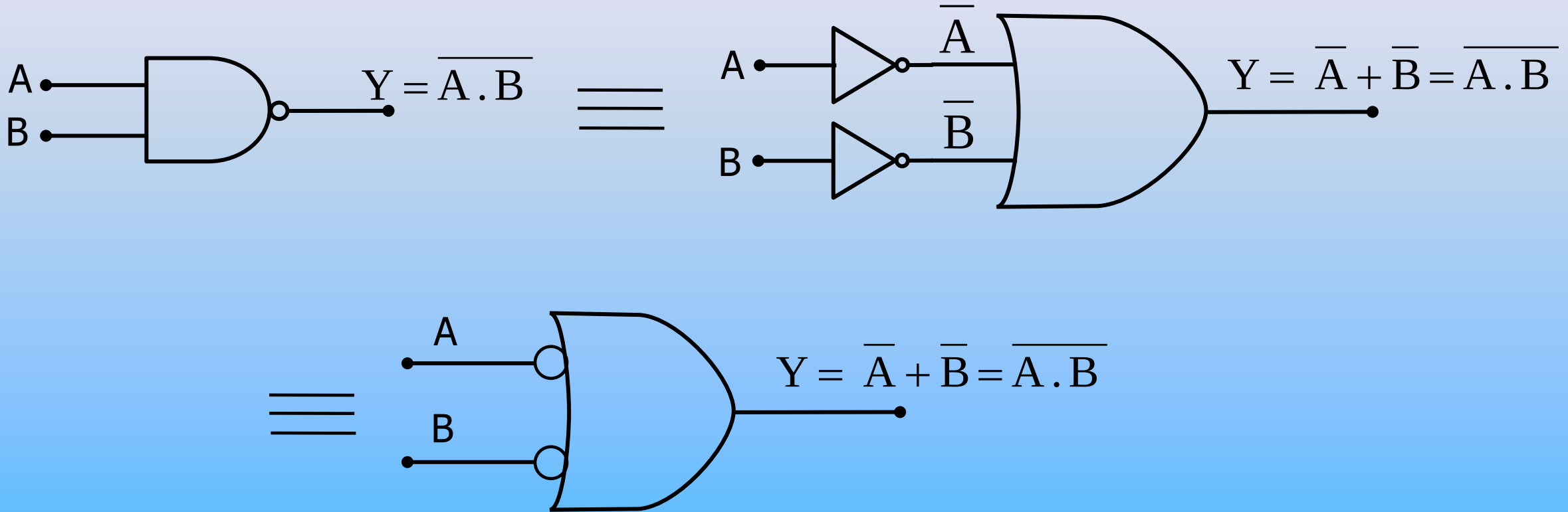
INPUT (A)	OUTPUT ($\bar{A}$)
0	1
1	0



NAND Gate: NAND is a contraction of the NOT-AND gates. It has two or more inputs and only one output, i.e. $Y = \overline{A \cdot B}$. When all inputs are HIGH, the output is LOW. If any one or both the inputs are LOW, then the output is HIGH.



Logic Symbol of NAND gate



NAND Gate $\equiv$ Bubbled OR gate

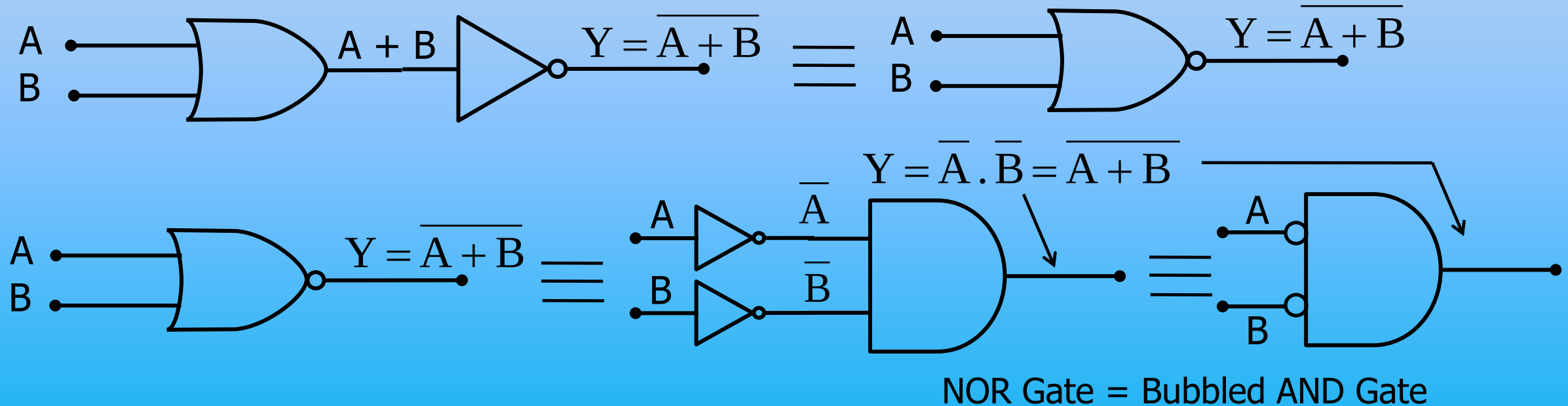


INPUTS		OUTPUT
A	B	$Y = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

TRUTH TABLE



NOR Gate: NOR is the contraction of NOT-OR gate. It has two or more inputs and only one output. i.e. $Y = \overline{A + B}$. The output is HIGH only when all inputs are LOW. If any one or both the inputs are HIGH, then the output is LOW.



NOR Gate = Bubbled AND Gate

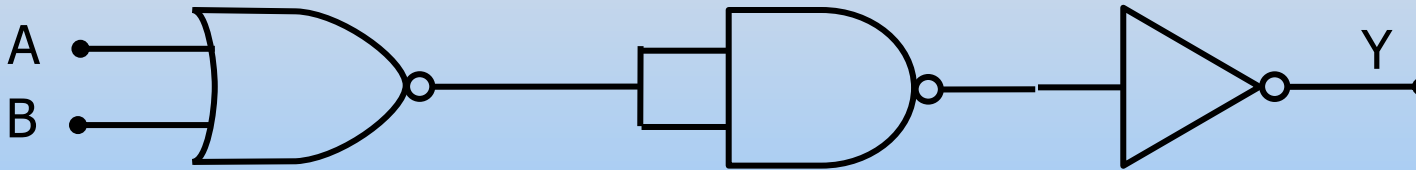


TRUTH TABLE OF NOR GATE

INPUTS		OUTPUTS
A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0



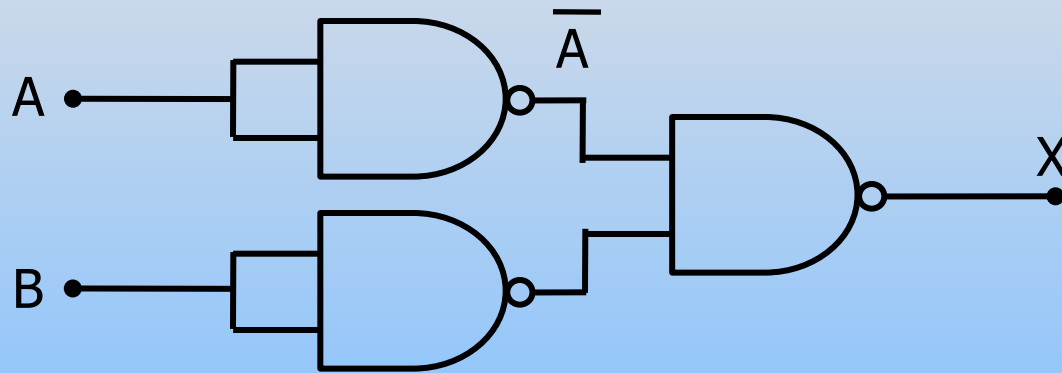
EXAMPLE 1: The given electrical network is equivalent to



- (a) OR gate (b) NOR gate
- (c) NOT gate (d) AND gate



EXAMPLE 2: The combination of gates shown below yields:



(a) XOR gate

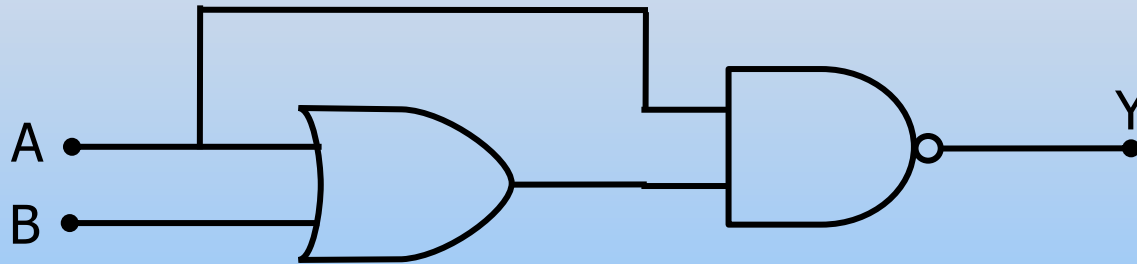
(b) NAND gate

(c) OR gate

(d) NOT gate



EXAMPLE 3: The truth table for the circuit given in fig. is:



(a)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

(b)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(c)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(d)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0



EXAMPLE 4: To get output 1 for the following circuit, the correct choice for the input is



(a) $A=0, B=1, C=0$

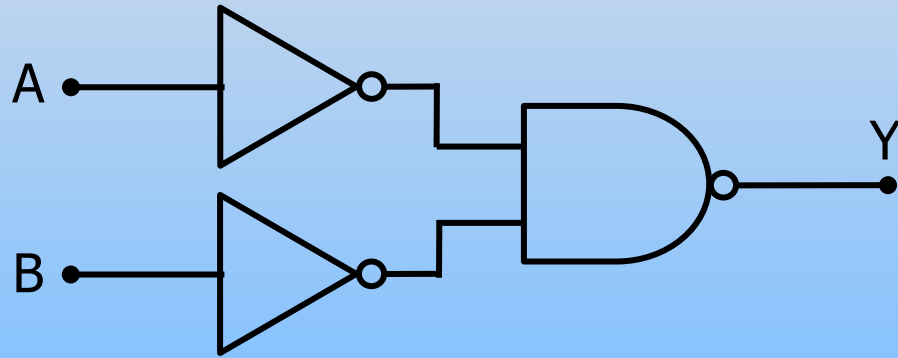
(b) $A=1, B=0, C=0$

(c) $A=1, B=1, C=0$

(d) $A=1, B=0, C=1$



EXAMPLE 5: The logic gate equivalent to the given logic circuit is:



(a) OR (b) AND

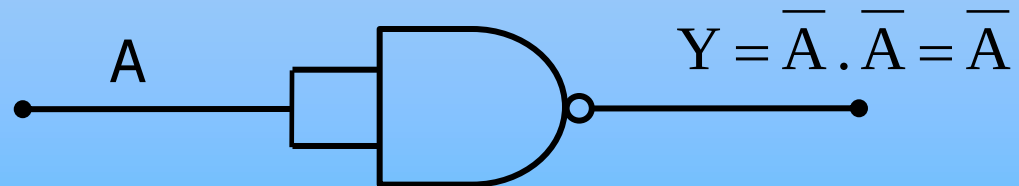
(c) NOR (d) NAND



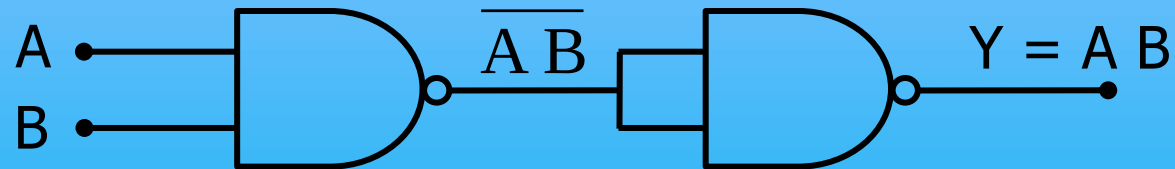
UNIVERSAL GATES / UNIVERSAL BUILDING BLOCK: NAND and NOR gates are called Universal gates or universal building blocks because both can be used to implement any gate like AND, OR and NOT gates or any combination of these gates.

REALISATION OF NOT, AND, OR, NOR GATES USING NAND GATES:

(1) Inverter or NOT gate

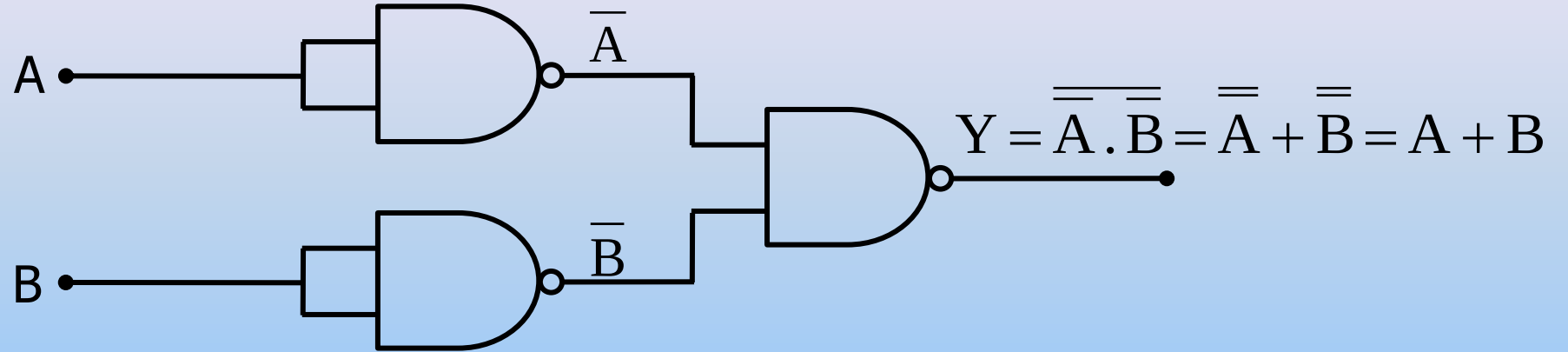


(2) AND Gate

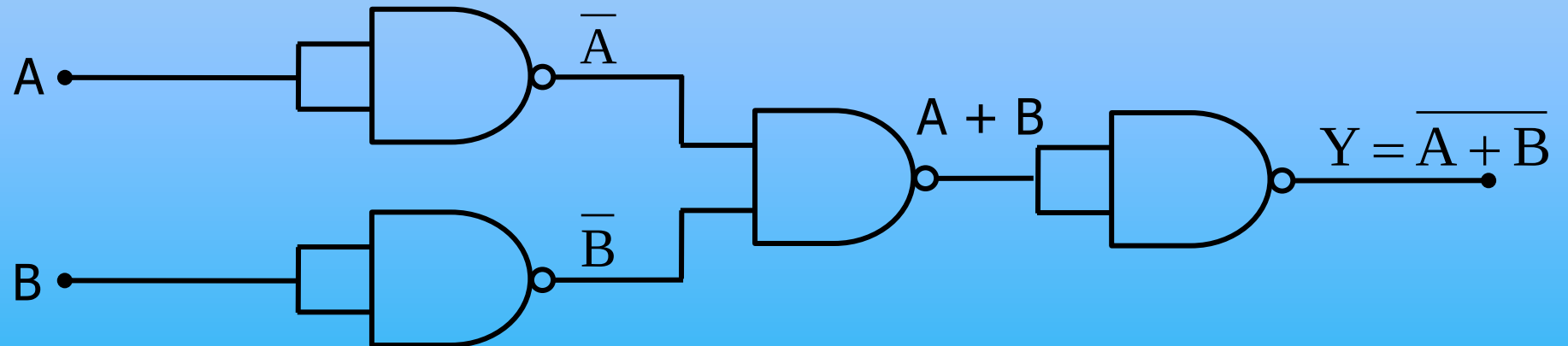




(3) OR Gate



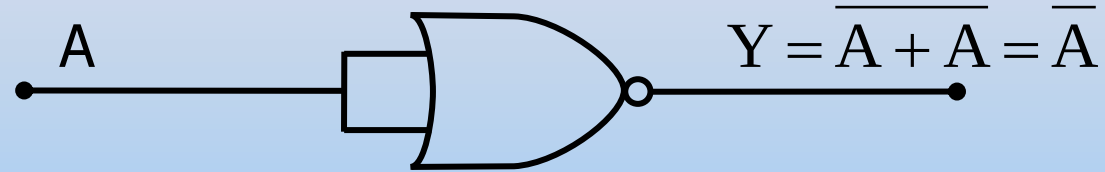
(4) NOR Gate



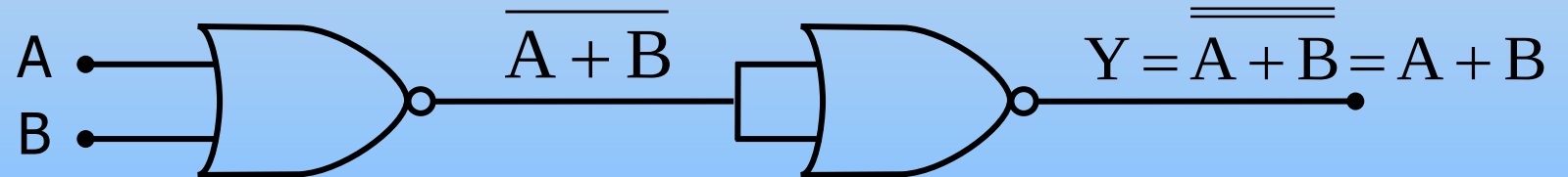


REALISATION OF NOT, AND, OR , NOR GATES USING NOR GATES:

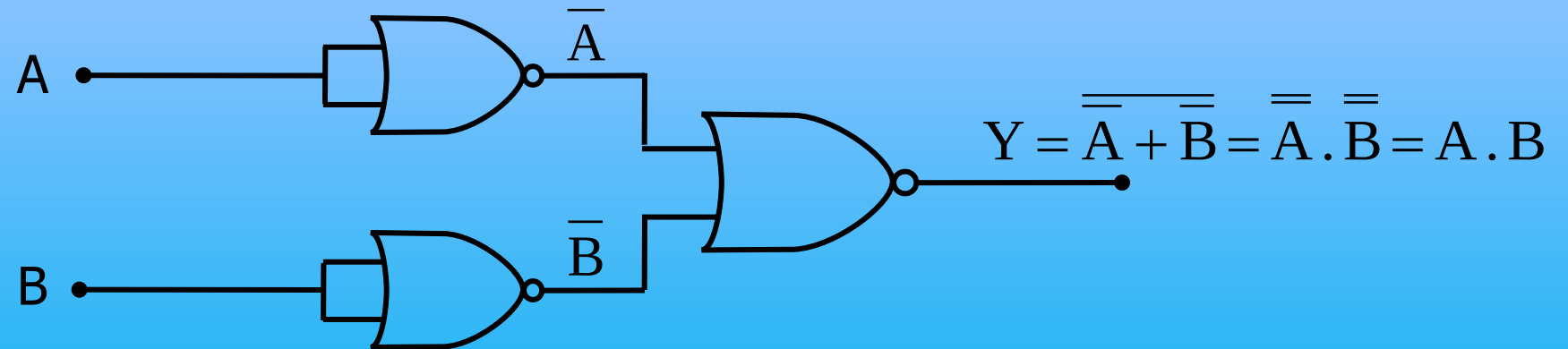
(1) NOT Gate



(2) OR Gate

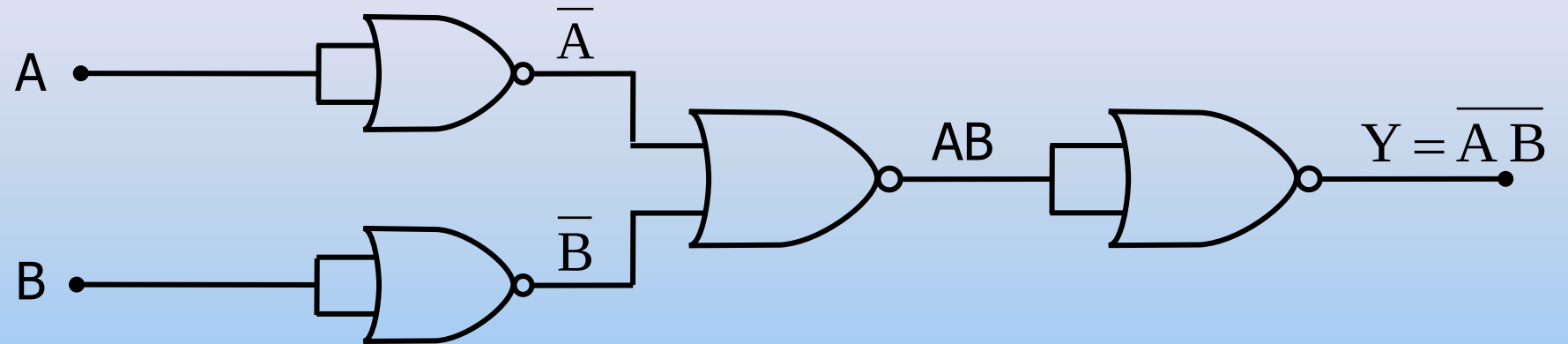


(3) AND Gate



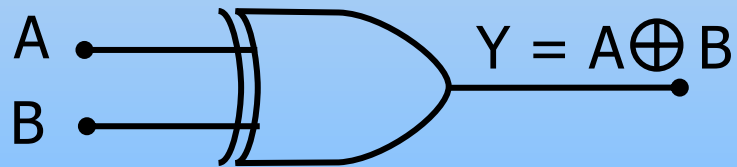


(3) NAND Gate





EXCLUSIVE OR GATE (EX-OR) GATE: An Exclusive OR gate is a gate with two or more inputs and one output. The output of a two input Ex-OR gate assumes a HIGH state if one and only one input assumes a HIGH state.



$$Y = \bar{A} B + A \bar{B} = A \oplus B$$

INPUT		OUTPUT
A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0



EXCLUSIVE NOR GATE (EX-NOR) GATE: An Exclusive NOR gate is an Ex-OR gate followed by an inverter. An exclusive-NOR gate has two or more inputs and one output. The output of a two input Ex-NOR gate assumes a HIGH state if both the inputs assumes the same logic state or have an even number of 1s and output is LOW when the inputs assume different logic states or have odd number of 1s.





INPUT		OUTPUT
A	B	$Y = \overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1



😊 **TAHINK YOU**

